Zayed Hasan

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EDUCATION

B.Sc. in Software Engineering 2021 – 2025
Islamic University of Technology (IUT), Gazipur, Bangladesh
CGPA: **3.74 / 4.00**
Thesis: *Pre-Processing the Prompt to Generate an Efficient Output for Unit Test Case Generation*

RESEARCH EXPERIENCE

Undergraduate Thesis — Islamic University of Technology Sep. 2024 – Sep. 2025
Abstract Syntax Tree Augmented Prompt Engineering for LLM-based Java Unit Test Generation
Supervisor: Jibon Naher, Dept. of CSE, IUT
Co-investigators: Lomatul Mahzabin, Shahrier Al Tanzim

Investigated an Abstract Syntax Tree (AST)-augmented prompt engineering framework for automated Java unit test generation via LLMs and static analysis, aiming to improve test coverage and correctness without expensive model fine-tuning.

- Selected 20 real-world, Maven-based Java projects from GitHub to experiment with realistic project structures and class-level dependencies.
- Designed a pipeline extracting AST from source code, serialized to structured JSON, then constructing prompts combining raw source, AST context, and a strict instruction set.
- Evaluated two prompt architectures — Baseline (raw source code and instructions only) vs. AST-based (augmented with structured Abstract Syntax Tree context) — across three LLMs: GPT-OSS-120B, Gemini-2.5-Pro, and Qwen-2.5-Coder-32B-Instruct; measured compilation success and statement/branch/line/method coverage.
- AST-based prompts achieved **62.09%** average line coverage and **49.81%** branch coverage vs. 50.82% and 37.27% for baseline; GPT compilation success reached **72.22%**, surpassing the TestBench state-of-the-art (50.7%).
- Identified and characterized a structural-vs-semantic hallucination trade-off: AST context reduced syntax errors (15.9% vs. 28.8%) and type errors (11.4% vs. 21.9%), while occasionally inducing hallucination-related errors, such as referencing non-existing class members or methods.

Undergraduate Researcher — Islamic University of Technology Sep. 2025
Commonsense Graph Reasoning for Interpreting Riddles
Tools: PyTorch, HuggingFace Transformers, spaCy, ConceptNet
Designed a hybrid NLP pipeline integrating commonsense knowledge graphs with transformer-based models to enhance both accuracy and explainability in riddle-solving — a challenging benchmark for figurative language understanding and multi-step commonsense reasoning.

- Combined commonsense knowledge graphs (ConceptNet, WordNet, NumberBatch) with transformer models, grounding predictions in structured external knowledge rather than pretrained weights alone.
- Fine-tuned DeBERTa-v3-base on RiddleSense (5,700 riddles), achieving **69.96%** dev accuracy — a **+15.8 pp** improvement over BERT-Base (54.16%) and competitive with UnifiedQA and ALBERT-XXL.
- Built an end-to-end workflow: key concept extraction, schema graph construction, contextual embeddings, and five-way classification via CrossEntropy loss and AdamW optimisation.
- Analysed graph–language fusion failure modes (graph noise, subgraph selection, structural-vs-semantic integration), situating findings against KagNet, RISCORE, and OpenCSR via literature review.

PROFESSIONAL EXPERIENCE

Software Engineer — Sheba Technologies Ltd. (resource-augmented to Breker Systems, USA) Nov. 2025 – present

- Work on the development of a software framework used for SoC/ASIC functional verification of semiconductor designs, developed in C/C++ within a Linux-based pipeline.
- Developed and maintained C/C++ components of this verification framework, integrating with test-generation and simulation workflows.
- Designed and validated instruction-level CPU tests covering RISC-V instructions and extensions, working directly with assembly-level logic, memory operations, and CPU register semantics.
- Worked on cache coherency verification, including the integration and validation of cache coherency tests for multi-core SoC systems.
- Contributed to processor and SoC-level functional verification, ensuring correctness of design behavior.

Software Engineering Intern — Logicri, Dhaka, Bangladesh Mar. 2025 – Jun. 2025

- Contributed to full-stack development and maintenance of a production-grade ERP system (ASP.NET Core, C#, Next.js, TypeScript, Node.js, Express.js) serving live client deployments.
- Applied clean architecture principles and RESTful API design patterns to ensure scalability and maintainability.

Software Engineering Intern — Streams Tech Ltd., Dhaka, Bangladesh Jun. 2024 – Oct. 2024

- Contributed to a fintech SaaS product (client: Spendplan, USA) using ASP.NET, Redis, and PostgreSQL.
- Implemented a backend Redis caching layer, reducing data retrieval latency and optimising server performance.
- Built a unit and integration testing suite; integrated automated tests into CI/CD pipelines via GitHub Actions.

SELECTED PROJECTS

AutoUnitTestGenerator *Github*
Python, LangChain, LLM integration (extension of undergraduate thesis)

- Automated pipeline for generating JUnit test classes from Java source using LLMs.
- Implements optional RAG with documentation retrieval and post-processing for compilable output.
- Automated Maven execution with coverage reporting for end-to-end test validation.

AI-Powered Personal Profile Q&A System *GitHub*
Python, RAG, LangChain, Ollama Feb. 2026

- Built a Retrieval-Augmented Generation (RAG) system using local embeddings and vector search.
- Answers queries strictly grounded in uploaded CV data, preventing hallucinated responses.
- Demonstrates applied understanding of local LLM inference and semantic retrieval pipelines.

Intelligent job finder agent *GitHub*
Python, Gemini API, LLM Agents, JobSpy Aug. 2026

- Built an autonomous agent that parses a CV into structured data and uses Gemini to plan search terms grounded in real experience and skills.
- Designed a closed-loop refinement cycle where the LLM evaluates each round's results and rewrites search terms accordingly.
- Demonstrates applied understanding of LLM-driven planning, evaluation, and iterative query refinement.

InternConnect *GitHub* *Live*
MongoDB, Express.js, React.js, Node.js, Netlify

- Developed a full-stack MERN web application to streamline internship management at IUT.
- Simplified student applications, tracking, and placement workflows for administrators and students.

TECHNICAL SKILLS

Languages	C, C++, C#, Java, Python, JavaScript/TypeScript, SQL, Assembly (RISC-V)
AI / ML	LLM integration, RAG, LangChain, Ollama, prompt engineering
Systems	RISC-V ISA, SoC/ASIC verification, Linux, low-level debugging
Web / Backend	ASP.NET Core, Node.js, Express.js, Spring Boot, Next.js, ReactJS
Databases	PostgreSQL, MySQL, MongoDB, Redis, Neo4j
Testing / DevOps	JUnit 5, JaCoCo, Jest, GitHub Actions (CI/CD), Maven, Git, Azure
Problem Solving	Codeforces, LeetCode

RELEVANT COURSEWORK

Data Structures & Algorithms • Software Testing & Quality Assurance • Software Verification & Validation • Software Maintenance • Software Design & Architectures • Computer Organization & Architecture • Microprocessor & Interfacing • Embedded Software Development • Server Programming • Database Management Systems • Artificial Intelligence • Design Patterns • Natural Language Processing • Software Security

REFERENCES

Jibon Naher
Lecturer, Department of Computer Science & Engineering
Islamic University of Technology (IUT), Gazipur, Bangladesh
(Undergraduate Thesis Supervisor)